

HEMANT KUMAR TYAGI

AI & Generative AI Engineer | Agentic Applications

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Professional Summary

AI & Generative AI Engineer with extensive experience in enterprise software development and hands-on engineering of AI-powered applications. Experience spans predictive maintenance and Generative AI solutions. Builds enterprise knowledge assistants using RAG and agentic workflows that connect conversational interfaces with HR and service-management systems. Brings expertise in Python, LangGraph, LangChain, FastAPI, and .NET Core APIs, supported by a strong foundation in backend development, databases, and cloud integration.

Technical Skills

Generative AI & Agentic Systems: LLM Application Development, RAG, Prompt Engineering, Embeddings, Semantic Search, Tool Calling, Agentic Workflows, Model Context Protocol (MCP), LangChain, LangGraph, LangSmith
AI/ML & Data: Predictive Modeling, Anomaly Detection, Time-Series Analysis, scikit-learn, pandas, NumPy
Backend & APIs: Python, FastAPI, C#, .NET Core, ASP.NET Web API, REST APIs
Cloud Platforms: Microsoft Azure (Azure OpenAI, Azure AI Search, Azure AI Foundry, App Service, Azure Functions, Blob Storage); AWS (Amazon EC2, Elastic Beanstalk, Amazon RDS); Google Cloud Platform
Databases & Vector Search: PostgreSQL, pgvector, Microsoft SQL Server, Azure SQL, Cosmos DB, MySQL, MongoDB
Frontend: React, Angular; AI-assisted Frontend Development

Professional Experience

Application Development Team Lead

February 2022 – Present

TELUS International

Noida, India

- Develop enterprise AI applications that combine Generative AI, knowledge retrieval, and API-driven automation to support business workflows.
- Build backend services using Python, FastAPI, and .NET Core APIs to integrate AI workflows with enterprise systems.

GenAI Knowledge Assistant with Agentic Workflows

Developed an enterprise knowledge assistant using a single agent with LangGraph and LangChain, integrating RAG and tools exposed through an MCP server.

- Implemented intent recognition, conversation context management, and tool selection within a single agent to handle knowledge queries and business actions.
- Integrated RAG to retrieve relevant enterprise documents and answer knowledge and HR policy questions with supporting references.
- Exposed leave application, ticket creation, and ticket-status retrieval as tools through an MCP server, with each tool invoking the corresponding enterprise API.
- Collected and validated required details for leave applications and ticket creation, and presented ticket progress, assignment, and recorded updates through conversational responses.

Technologies: Python, LangGraph, LangChain, LangSmith, FastAPI, MCP, Azure OpenAI, Azure AI Search, Azure SQL Server.

AI Predictive Maintenance & Fleet Intelligence

- Developed predictive maintenance and anomaly-detection capabilities for heavy construction equipment fleets, including backhoe loaders, excavators, dozers, articulated trucks, off-highway trucks, and compactors.
- Built Python-based data-processing and feature-engineering pipelines to prepare equipment telemetry and sensor data for predictive modeling.
- Integrated model predictions and equipment-health insights into .NET Core APIs and operational dashboards to support maintenance planning and fleet monitoring.

Technologies: Python, scikit-learn, pandas, NumPy, .NET Core APIs, PostgreSQL.

Professional Experience (Continued)

Assistant Manager, Software Development

May 2019 – February 2022

Teleperformance

Gurugram, India

MyHiring HRMS Project

Worked as a hands-on individual contributor developing ASP.NET MVC and .NET APIs, SQL Server components, and HRMS enhancements for employee onboarding, candidate registration, and employee information workflows.

- Integrated third-party APIs and investigated and resolved production defects.
- Developed candidate registration and bulk candidate upload functionality. Integrated Vue.js frontend workflows with Consultant Hiring backend APIs.
- Enhanced the MyHiring portal with electronic onboarding and candidate registration for hiring vendors, and developed Employee Detail Form (EDF) and Personal Detail Form (PDF) editing functionality integrated with MyHiring Core APIs. Supported application security and code-quality testing with WAPT, HCL AppScan, and SonarQube.

Technologies: ASP.NET MVC, ASP.NET Core, Microsoft SQL Server.

Software Developer

January 2018 – April 2019

Simplex Control Equipments Co.

Dehradun, India

- Developed industrial automation applications using C# and ASP.NET MVC, integrating PLC/SCADA systems and processing IoT sensor data.
- Built real-time monitoring, threshold-based alerts, and SQL Server-backed reporting for equipment operations.

Software Developer

May 2014 – January 2018

Himja Software

Dehradun, India

- Developed business web applications using ASP.NET MVC, C#, and SQL Server, translating client requirements into application features.
- Built responsive interfaces using JavaScript, jQuery, and Bootstrap, and maintained existing application modules.

Software Developer

September 2011 – April 2014

Dreamz Infinite

Delhi, India

- Developed data-driven web applications using ASP.NET MVC, C#, and SQL Server.
- Created stored procedures and database queries, and contributed to application development, deployment, and support.

Education

Bachelor of Computer Applications (BCA)

2008 – 2011

Uttaranchal Institute of Management

Dehradun, India